Guidelines for Overcoming the Practical Limitations for the Fabrication of THz Sources with GaN Planar Gunn Diodes

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Abstract

Gunn oscillations are expected to appear in GaN at high electric fields due to its negative differential mobility, but they have never been observed experimentally due to the large voltages to be applied. We propose a way to avoid such practical problems by fabricating planar Gunn diodes with shaped nanochannels which allow to reduce the total DC current and focus the electric field in regions far from the contacts.

Keywords: GaN, Gunn diode, THz sources

Introduction

The bottleneck for the broad development of practical applications exploiting the THz range of the electromagnetic spectrum is the lack of room-temperature, continuous-wave, compact, mass producible, tuneable and powerful sources. However, in spite of the technical difficulties, the field of Terahertz Science and Technology is gaining international interest due to its broad potential applications, ranging from ultra-high speed wireless transmission systems to medical diagnostic, industrial quality control, security-screening tools, THz astronomy, high speed communications, or pharmacology [1]. Photonic and electronic approaches compete for conquering the "THz gap" (see for example the topical review [2]), the latter having the advantage of being less bulky and simpler to use. The electronic devices having exhibited the highest oscillation frequencies (at over 1 THz [3]) are resonant tunnelling diodes (RTDs), but, due to their small size, the output power is very low at high frequencies (even if recent technical advances have allowed RTD sources to provide power in excess of 0.7 mW at 1 THz). Transistor-based MMICs have also been demonstrated to be suitable for fabricating high frequency oscillators [4] with a good level of output power (10 mW@300 GHz), but the fabrication process is extremely complex. Frequency multiplication with GaAs Schottky barrier diodes (SBDs) [5] based on fundamental oscillators such as TUNNET, IMPATT or transferred-electron devices (also called Gunn diodes [6]), which generate output power taking advantage of a negative differential resistance (NDR) at the frequency of interest, is also a good alternative. In particular, the simple, compact and flexible planar Gunn diode (PGD) architecture is gaining popularity in the last years. GaAs- and InGaAs-based PGDs have demonstrated fundamental oscillation frequencies of 120 GHz (the highest for a GaAs-based Gunn diode [7]) and above 300 GHz [8], respectively. Gunn diode technology is the presently preferred approach due to its compactness, tunability and power level.

However, in spite of these remarkable results, a widespread availability of THz sources is still lacking for accomplishing mass-market THz applications. A broad access to room-temperature, high-power, tunable, compact and (hopefully) low-cost sub-THz sources, can be a starting point to grant an extensive development of ultrafast communications, security detection scanners, radars for self-guided vehicles, etc.

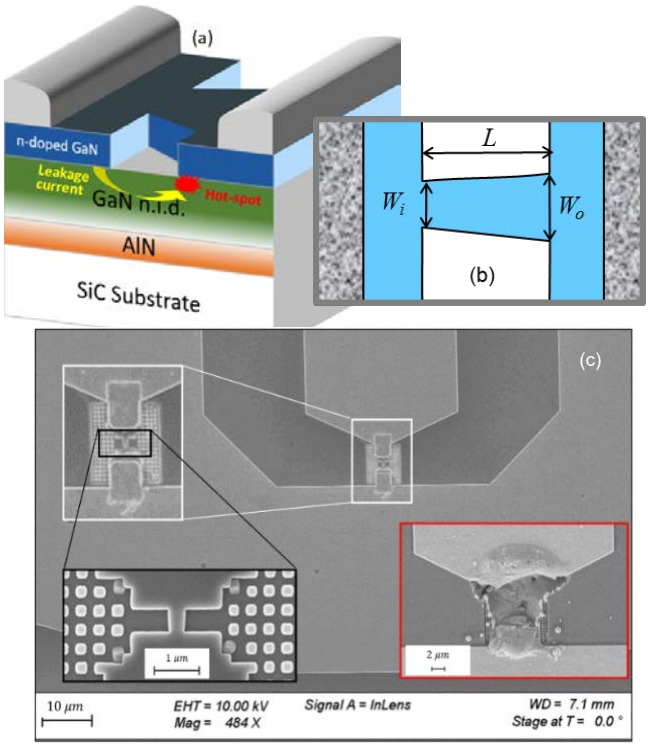


Fig. 1: (a) 3D geometry of the geometrically shaped Gunn diodes based on doped GaN and (b) top-view sketch of a single channel device. (c) SEM picture of a fabricated diode formed of a tapered channel with W_i=300 nm, W_o=350 nm and L=0.5 μm. The bottom right picture shows a diode after its catastrophic failure.

GaN Planar Gunn Diodes

The development of GaN technology has made significant inroads into high-power and high-frequency applications with respect to other semiconductors and has emerged as one

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of the most promising materials for high power applications at microwave frequencies. In the last years, many efforts have been dedicated to the intent of fabricating high-power high-frequency Gunn diodes based on GaN [9, 10, 11]. Our approach is based on a planar fin-type Gunn diode topology based on tapered-channel geometry (or geometrically-shaped PGDs, GS-PGDs), Figure 1(a), which can be used to achieve THz sources by improving the efficiency of planar Gunn diodes due to a dual action. (i) The electrical field is well focused at the cathode side of the channel, so that the threshold voltage needed for the onset of the oscillations is reduced; the narrow part of the channel acts as a notch that also allows a better control of the oscillation frequency. And (ii) the heat dissipation can be efficiently managed by an appropriate design of the channels of GS-PSDs and its separation. As a consequence of a lower applied voltage and weakened thermal effects, the device breakdown is avoided, so that continuous wave operation can be reached. Higher output power and frequencies (in excess of 300 GHz) are expected to be reached with GaN Gunn diodes due to its high saturation velocity and high breakdown field [12]. However, no evidence of Gunn oscillations was observed in the fabricated devices [13,14], even if simulations predict the appearance of Gunn oscillations in GaN diodes [12].

We have recently identified the origin of such negative outcome. In the case of diodes based on the AlGaN/GaN heterostructures, the space charge created by the Gunn domains is associated with the positive surface charge located at the heterojunction, developing within an undoped GaN layer, so that it cannot change abruptly. This is confirmed by the indirect evidences of current oscillations in diodes based on doped GaN layers [9], together with the recent observation of NDR in vertical diodes [10], which allows us to envisage the practical feasibility of high-power high-frequency GaN Gunn diodes. That is why we still rely on the GaN PGDs approach, based on the geometrical shaping of channels, but including an important novelty: the use a doped GaN active layer instead of the typical AlGaN/GaN epilayers employed so far. Simulations show that by increasing the doping of the active layer, the Debye length is much reduced, so that the formation of the Gunn domain is promoted by the presence of a higher electric field, which appears very fast in the close vicinity of a space charge outbreak. Therefore, the main benefit of the use of a doped GaN active layer is the enhanced strength of the local electric field when a charge imbalance appears, with the additional advantage that it can be modulated by means of the doping level.

The GS-PGDs of Fig. 1 were fabricated on GaN epilayers, with optimally chosen doping levels (in the range N_D=5-10×10^{19}cm^{-3}) and thickness (in the 125-200 nm range), grown on the top of an undoped GaN buffer and a AlN nucleation layer using the process as described in [15,16]. Not only the thickness and doping level of the active layer, but also the geometries of the devices were optimized by means of Monte Carlo simulations (with a careful consideration of surface effects [11]) in order to enhance the possibility of generating Gunn oscillations, trying to avoid self-heating effects and considering the constraints imposed by the technological process for the fabrication of the devices (which imposes a limit to the thickness of the active layer of about 200 nm). Unfortunately, a catastrophic breakdown of the fabricated devices for applied voltages much lower than expected is observed, regardless of the epilayer structure and device geometry, Fig. 1(c).

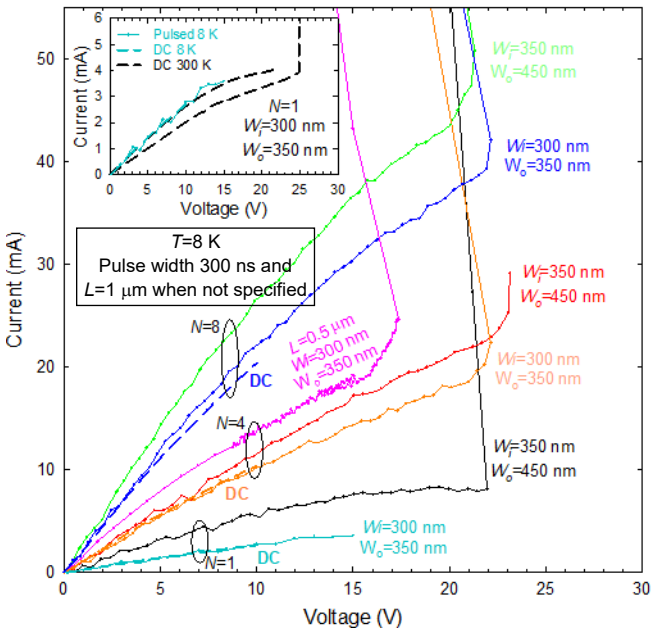


Fig. 2: Pulsed measurements performed at T=8 K using 300 ns-width pulses (1% duty cycle) in order to avoid as much as possible self-heating effects (thin dotted lines). In order to demonstrate that self-heating effects are almost negligible, the DC measurements for the diodes with W_i=300 nm and W_o=350 nm are also plotted with thick dashed lines. The inset shows the I-V curves obtained with DC measurements at T=8 and 300 K (providing similar values, as expected from the nearly temperature-independent values of mobility and sheet electron density) for a single channel diode (N=1) with L=1 μm, W_i=300 nm and W_o=350 nm, compared with the pulsed results at T=8 K. Two consecutive sweeps at 300 K applying voltages up to 22 and 25 V are represented (failure was observed in the second sweep).

By means of a detailed characterization of the fabricated GS-PGDs (trying to overcome the catastrophic breakdown due to thermal effects often experienced at room temperature in this type of devices [10]), we have demonstrated that the failure mechanism is not linked to thermal heating. In Fig. 2 it can be clearly observed that low-temperature pulsed bias conditions are not able to avoid the failure of the devices at voltages below 25 V. Indeed, the breakdown voltage is similar to that obtained under DC bias conditions and at room temperature for N=1 (see inset), as expected since self-heating effects are negligible. Indeed, due to the low current level provided by the devices, a current decrease between DC and pulsed measurements is observed only for dissipated powers above ~0.1-0.2 W, typically reached only for N=8.

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The most interesting observation is that the device failure takes place when the electric field within the channel reaches about 200 kV/cm (for devices with both L=1.0 and 0.5 µm), just the value for which the peak velocity is found and the NDR region starts. As a consequence, we attribute the failure of the GS-PGDs to a mechanism related to the high electric fields appearing within the channels. 3D simulations (using Silvaco ATLAS) show the location of an electric field hot spot (with values well above the MV/cm range) at the bottom corner of the trenches in the anode side, see Fig. 1(a), which leads to the onset of an avalanche process started by the leaked electrons traveling below the isolating trenches, promoted by the formation of Gunn domains.

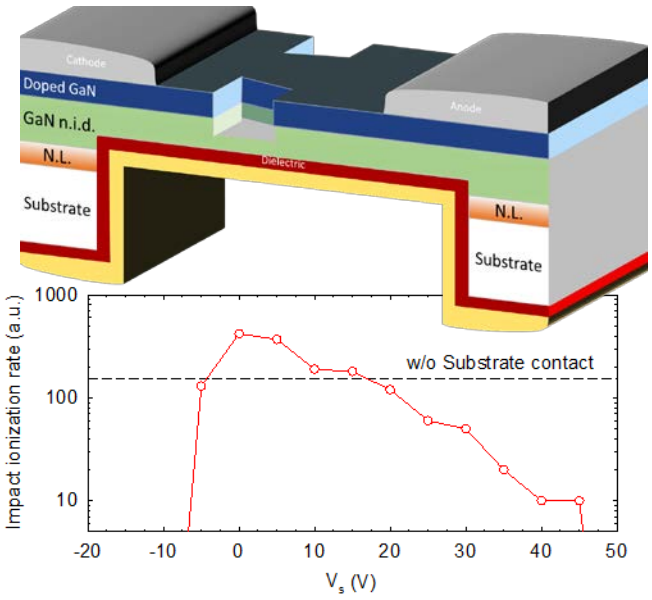


Fig. 3: 3D geometry of the proposed 3T-GS-PGD with substrate contact and impact ionization rate *vs*. V_S for an anode-cathode bias of V=30 V compared with those obtained for a GS-PGD (in absence of the substrate contact, dashed horizontal line).

Monte Carlo simulations show that the avalanche process is initiated by the electrons passing below the isolation trenches shown in Fig. 1(a). Thus, improving the isolation by realizing a deeper etch could be the way to reduce the effect of impact ionization processes and thus avoid the failure of the diodes. A value of the depth of the trenches around 250 nm (100 nm deeper than the active layer thickness) would be enough to almost completely suppress the leakage current and thus avoid the breakdown. However, such deep trenches are not simple to achieve without degrading the definition of the channel shape. That is why we propose another possibility to overcome the practical limitations for achieving Gunn oscillations: introduce a bottom contact to the GS-PGD (leading to a 3T-GS-PGD) in order to suppress the impact ionization mechanisms by applying an appropriate voltage V_S. For that sake, it is necessary to realize a bottom recess in order to approach the back side of the active region, and to deposit a thin dielectric in order to avoid the substrate current. As shown in Fig. 3, the substrate terminal acts in two different ways: (i) suppressing the electron leakage flow below the trenches by applying V_S<0 or (ii) decreasing the electric field at the hot-spot at the anode corner of the trenches by applying V_S>0 (with a value similar to that applied to the anode, which can be applied by simply short-circuiting both terminals thus avoiding another DC connection). In both cases the number of impact ionization mechanisms are much suppressed with respect to those obtained in a GS-PGD with the same geometry. More results of simulations of this new structure will be presented at the conference showing how the topology of the devices can be optimized.

Acknowledgments

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